



EcoSphere: ESG Management Platform

1. Background

Environmental, Social and Governance (ESG) has become a critical aspect of modern businesses. Organizations are expected to monitor carbon emissions, promote employee well-being, and maintain governance compliance. While many ERP systems collect operational data, ESG reporting is often manual, disconnected, and difficult to monitor in real time.

EcoSphere aims to integrate ESG directly into day-to-day ERP operations by measuring sustainability metrics, encouraging employee participation, and providing meaningful reports for management.

2. Challenge Statement

Build an ESG Management Platform that enables organizations to measure, manage and improve their Environmental, Social and Governance performance.

The platform should integrate operational data, employee participation and compliance activities into a unified dashboard while encouraging sustainability through gamification.

3. Core Modules

- **Environmental:** Carbon accounting, emission factors, sustainability goals and carbon reports
- **Social:** CSR activities, employee participation, diversity metrics and engagement
- **Governance:** Policies, audits, compliance tracking and governance reports
- **Gamification:** Challenges, badges, XP, rewards and leaderboards

4. Suggested Data Model

The application consists of Master Data and Transactional Data.

Master Data

Model	Purpose	Key Fields
Department	Organizational hierarchy and ESG ownership	Name, Code, Head, Parent Department, Employee Count, Status
Category	Shared category values used across Social and Gamification modules (e.g., CSR Activity Category, Challenge Category)	Name, Type (CSR Activity / Challenge), Status
Emission Factor	Carbon values used during calculations	-
Product ESG Profile	ESG information linked to products	-
Environmental Goal	Sustainability targets	-
ESG Policy	Governance policies	-
Badge	Employee achievements	Name, Description, Unlock Rule, Icon
Reward	Redeemable incentives	Name, Description, Points Required, Stock, Status

Transactional Data

Model	Purpose	Key Fields
Carbon Transaction	Stores calculated emissions from ERP operations	-
CSR Activity	Social initiatives organized by the company	-
Employee Participation	Tracks employee involvement in CSR Activities only	Employee, Activity, Proof, Approval Status, Points Earned, Completion Date
Challenge	Sustainability challenges	Title, Category, Description, XP, Difficulty, Evidence Required, Deadline, Status (Draft / Active / Under Review / Completed / Archived)
Challenge Participation	Tracks employee progress within Challenges only	Challenge, Employee, Progress, Proof, Approval, XP Awarded
Policy Acknowledgement	Employee policy acceptance	-
Audit	Governance audits	-
Compliance Issue	Governance violations	Audit, Severity, Description, Owner, Due Date , Status
Department Score	Aggregated ESG performance per department	Department, Environmental Score, Social Score, Governance Score, Total Score

5. Business Workflow

None

Master Configuration

|



Departments · Categories · Emission Factors · Products

Goals · Policies · Challenges

|



Daily Business Operations

(Purchase · Manufacturing · Expenses · Fleet)

|



Carbon Transactions

|



Employee Participation (CSR) · Challenge Participation

Policy Acknowledgements · Audits

|



6. Expected Features

Environmental

- Configure Emission Factors
- Calculate Carbon Emissions
- Department Carbon Tracking
- Sustainability Goals
- Environmental Dashboard

Social

- CSR Activities
- Employee Participation
- Diversity Metrics
- Training Completion

Governance

- ESG Policies
- Policy Acknowledgements
- Audits
- Compliance Issues

Gamification

- Challenges (with full lifecycle: Draft → Active → Under Review → Completed, or Archived at any point)
- XP
- Badges (auto-awarded when an employee's XP or completed-challenge count satisfies the Badge's Unlock Rule)
- Rewards (redeemable using earned XP/Points - see Section 8)
- Leaderboards

Settings & Administration

- Departments management
- Category management
- ESG Configuration (see business rules below)
- **Notification Settings** (see Section 8)

7. Reports

The platform should generate:

- Environmental Report
- Social Report
- Governance Report
- ESG Summary Report
- **Custom Report Builder** - lets users build a report by combining filters below and export it (PDF / Excel / CSV)

Each report should support filtering by:

- Department
- Date Range
- Module
- Employee
- Challenge
- ESG Category

8. Core Configuration & Business Rules

The following are **in scope**, not optional, since they directly support core modules:

- **Reward Redemption:** Employees can redeem earned Points/XP for a Reward from the catalog, subject to stock availability. Redeeming a Reward deducts the corresponding Points from the employee's balance.
- **Notification System:** The platform sends notifications (in-app and/or email) for at least: new compliance issue raised, CSR/Challenge approval decisions, policy acknowledgement reminders, and badge unlocks. Configurable via Settings → Notification Settings.
- **Auto Emission Calculation:** When enabled (Settings toggle), Carbon Transactions are calculated automatically from linked Purchase/Manufacturing/Expense/Fleet records using the relevant Emission Factor - no manual entry required.
- **Evidence Requirement:** When enabled (Settings toggle), CSR Activity participation cannot be marked Approved without an attached proof file.
- **Badge Auto-Award:** When enabled (Settings toggle), a Badge is automatically assigned to an employee the moment their XP, completed-challenge count, or other tracked metric satisfies that Badge's Unlock Rule - no manual admin action required.
- **Compliance Issue Ownership:** Every Compliance Issue must have an assigned Owner and a Due Date; issues that pass their Due Date while still Open should be flagged (feeds the Notification System above).

9. Bonus Ideas (Optional)

Participants are encouraged to explore additional features such as:

- Department ESG rankings
- Smart dashboard visualizations
- Mobile-responsive interface

Mockup: <https://link.excalidraw.com/l/65VNwvy7c4X/2m6lz9Ln4>